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Illuminating
ENGINEERING SOCIETY

RECOMMENDED PRACTICE: ECONOMIC ANALYSIS OF LIGHTING

AN AMERICAN NATIONAL STANDARD

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IES Economics Committee

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AMERICAN NATIONAL STANDARD

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1.1 Introduction:

Good lighting should be responsive to the needs of the user. Among those needs are the aesthetic and the visual, as admitted in the oft-quoted "lighting is both a science and an art." But the user also has economic needs. In fact, it is the economic needs that often drive the decision-making process when lighting systems are designed and purchased. Unfortunately, because they frequently control the final decision, economic concerns are often thought of as the antagonists of aesthetic and visual concerns. The lighting professional will tend to draw up a list of system desiderata, then heave a large sigh of resignation and say "but the budget won't allow it…"

This Recommended Practice is written from the point of view that "economic analysis" is not the same as "how to beat the budget." Rather than considering economic analysis as the antithesis of engineering or artistic analysis, it should be thought of as subsuming these other needs. A couple of generic examples illustrate this. When a worker's vision is impaired by reflected glare, reduced productivity is an economic consequence. A decision to improve the lighting would be based on the economic needs of the owner. When the lighting of an office building atrium fails to complement the architecture of the space, the rental value would fail to achieve its potential. Again, all decision to improve the lighting is an economic decision. Thus, when a competent lighting professional takes care of economic needs in conjunction with artistic, engineering, and other needs, it increases the likelihood a project will have success and longevity. Financial considerations as demonstrated through an accurate lighting financial analysis is important, but other elements such as aesthetics, human visual performance resulting from a lighting system appropriate to a given task, and other considerations involved in lighting for the human and natural environment are of equal importance.

Note: Parts of this document are adapted from chapter 14 of *Lighting for Energy Efficient Luminous Environments,* by Helms and Belcher.[1] Used by permission.

1.2 Scope

This Recommended Practice (RP) will help answer many types of lighting economic questions. It provides a framework for selecting from a group of competing lighting designs. It gives insight into the question of when a system under consideration will "pay off." It can help the lighting professional make energy conservation decisions. Most importantly, it provides methods for gauging the profitability of a capital investment in a lighting system, which can be objectively compared to other competing capital investments.

This RP begins with a discussion of the second-level methods, concentrating on LCCBA. This is followed by sections on sensitivity analysis and benefit analysis. Finally, because the lighting practitioner is likely to encounter the first-level methods in practice, these are also covered. In discussing the first-level methods, emphasis is placed on their shortcomings so that the lighting professional can understand why their use is not encouraged.

2.0 The Role of Economic Analysis in Lighting Design

2.1 The Importance of Making Wise Economic Decisions in Selecting Lighting Systems

Lighting economics may not be the most exciting subject for the lighting professional to master, but it is one of the most important. Reducing energy use is a noble pursuit for many reasons. Preserving natural resources through conserving fossil fuels is good for the future. Limiting air pollution and reducing carbon dioxide and other greenhouse gases helps keep the environment healthier. If demand for electricity is steadily growing through economic growth in North America and the rest of the world, by reducing the demand for energy needed to power lighting systems, more resources can be made available for other uses. Reducing consumption while still providing adequate levels and quality of illumination when and where needed makes good sense for any individual, company, government, or institution paying its own energy bill.

The increased cost associated with buying a more efficient lighting system can be looked at as an investment like any other kind of investment. The returns can be very high, as much as 60% per year or more, but with one significant difference. The risk for an investment in lighting is relatively low because the resulting cash flows for lighting are relatively predictable. In general, investments with lower financial risk, such as lighting upgrades, are analyzed with interest rates that can be several percentage points lower than more-risky capital investments.

With the dramatic growth in solid state lighting, including LED systems, the use of economic tools and rigorous financial analysis is more important than ever. Selecting a new lighting system with a less familiar technology is an important and difficult task for the lighting professional. The designer needs to make sure that the proposed lighting system meets or exceeds IES recommendations for the application, complies with energy code power limits and control requirements, and is able to meet extra targeted goals for the building, such as those from LEED, Advanced Energy Design Guides, or other local or project standards. But at the most practical level, any new lighting equipment selection needs to make sense economically. The days of specifying an LED lighting system just because it is “green” are largely in the past. Most owners and architects want to help the environment and reduce energy waste, but they also want good lighting at an affordable overall cost that makes sense. Owners often will not mind paying more for an improved system, but they want to get a return back from this increased cost. Solid financial analysis tools are essential to prove that such a system can be a better value when compared to traditional sources, i.e., incandescent, halogen, fluorescent, or HID.

2.2 Types of Economic Analysis

Many metrics and techniques for economic analysis have been proposed over the years. These methods can be classified into two categories: *first-level analysis methods* (see **Section 6**), and *second-level analysis methods* (see **Section 3**).

First-level analysis methods include:

- Cost of light
- Simple payback
- Simple rate of return

Second-level analysis methods include:

- Life cycle cost/benefit analysis
- Savings investment ratio
- Internal rate of return
- Net present value

The first-level methods are attractive due to their simplicity. However, they can yield misleading results and therefore should be used for quick estimates involving short payback periods only. Of the second-level methods, life cycle cost/benefit analysis (LCCBA) has emerged as the most robust method and the one that is accepted by experts in managerial economics from all industries. Accordingly, LCCBA is the economic analysis method recommended by the IES. This RP begins with a discussion of the second-level methods, concentrating on LCCBA. Because the lighting practitioner is likely to encounter the first-level methods in practice, these are also covered. In discussing the first-level methods, emphasis is placed on their shortcomings so that the lighting professional can understand why their use is not encouraged.

3.0 Second-Level Analysis Methods

The distinguishing feature of the second-level methods is also the feature that makes them superior to other approaches: they include the time *value of money*. The fact that time and money are interrelated can be easily demonstrated by posing the question “which would you rather have, $1,000 today, or $1,000 a year from now?” Even if one doesn’t care about when the money will be spent, the answer is obvious; the money is preferred now. Having the $1,000 now allows investing in interest bearing instruments, which will yield more than $1,000 a year from now. Alternatively, the $1,000 today can be used to purchase something from which the benefits can be enjoyed for an extra year (as compared to receiving the $1,000 a year from now).

This time value of money can be quantified. One way to do this is to pose a series of questions such as:

- Which is preferred? $1,000 today or $1,010 a year from now?
- Which is preferred? $1,000 today or $1,020 a year from now?
- Which is preferred? $1,000 today or $1,030 a year from now?

At some point it will become impossible to choose between the thousand dollars today and the future sum. A person might be equally willing to receive $1,000 today or, say, $1,100 one year from now. These two values are called *time equivalents*. From them an interest rate is inferred. In this example, the interest rate that equates the two time-equivalent values is 10 percent, and the time value of money in this case is said to be 10 percent per year.

Another term for this interest rate is the *opportunity rate of capital*, or simply, the *opportunity rate*. A person is willing to forgo the opportunity of receiving a sum of money today in exchange for the assurance of receiving an amount in the future equal to or greater than that sum, augmented by, or compounded at, the opportunity rate. Similarly, a person whose opportunity rate is 10 percent is willing to spend $1,000 today in order to avoid the need to spend $1,100 or more one year from now.

This concept can be used to allow the lighting professional to determine how much can be profitably invested in a lighting system today in order to incur the energy and maintenance costs of specified amounts in the future. It allows the comparison, in other words, of “apples and oranges”—dollars today vs. dollars tomorrow. This is the basis of all second-level methods. All economic events in the life of a lighting system (including initial cost, maintenance, energy cost, and salvage value) are converted into their value today, or *present value*, using the principle of time equivalence. The benefits are totaled and compared with the sum of the costs. If the benefits are greater, the system should be purchased. If the costs are greater, it would be unprofitable to purchase the system.

In calculating equivalences, four elements are involved:

- P: The amount of money being considered today, called the *present value*; in mortgage lending, this value is called the *principal*
- i: The *interest rate*, expressed as a percent or decimal fraction
- n: The period of *time* being considered
- F_n: The value (amount of money) being considered in the future (after time period n), called the *future value*

In the initial discussion example, P = \$1,000, n = 1 year, F_1 = \$1,100, and i was calculated to be 10%. A general formula involving these four variables can be derived such that, given any three, the fourth can always be found. Consider investing \$1,000 for three years at 10% interest, compounded annually. *Compounding* means that interest is paid on interest as well as on the principal. At the end of one year, the initial \$1,000 investment has grown to the future value (F_1):

$$F_1 = 1{,}000 + (1{,}000)(0.10) = \$1{,}100$$

After two years the future value (F_2) is:

$$F_2 = F_1 + (F_1)(0.10)$$

Therefore:

$$F_2 = 1{,}100 + (1{,}100)(0.10) = \$1{,}210$$

And after three years the future value (F_3) is:

$$F_3 = (F_2) + (F_2)(0.10)$$

Therefore:

$$F_3 = 1{,}210 + (1{,}210)(0.10) = \$1{,}331$$

The result is that \$1,331 three years hence is equivalent to \$1,000 today, at 10 percent interest compounded annually. This example can be generalized to n years at interest rate i (where i is a decimal fraction):
At the end of one year:

$$F_1 = P + (P)(i) = (P)(1 + i)$$

After two years:

$$F_2 = F_1 + (F_1)(\mathrm{i}) = (P)(1 + i) + (P)(1 + i)(i)\ ,$$

which, after factoring out $(P)(1 + i)$, yields:

$$F_2 = (P)(1 + i)(1 + i) = (P)(1 + i)^2$$

After three years:

$$F_3 = F_2 + (F_2)(\mathrm{i}) = (P)(1 + i)^2 + (P)(1+i)^2\ (i)\ ,$$

which, after factoring out $(P)(1 + i)^2$ yields:

$$F_3 = (P)(1 + i)^2\ (1 + \mathrm{i}) = (\mathrm{P})(1 + \mathrm{i})^3$$

Generalized for n years this expression becomes:

$$F_n = (P)(1 + i)^n \qquad \textbf{(3-1)}$$

Equation 3-1 yields the *future equivalent value* of any present value for given conditions of interest (i) and time (period n). By similar reasoning, other useful equivalents can be found. The terms used in the following equations are:

P = present value (cost in today's dollars)

F_n = future value, n years from now

A = uniform annual cost (such as yearly expenditure or yearly savings)

I = opportunity rate or interest

n = period (years) until the event (future value F_n) occurs

To find a present value, given a future value, an interest rate, and a period, **Equation 3-2** is used:

$$P = \frac{F_n}{(1+i)^n} \qquad \textbf{(3-2)}$$

To find present value, given an annual cost, or series of annual costs, an interest rate, and a period, **Equation 3-3** is used:

$$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] \tag{3-3}$$

To find a future value, given an annual cost, an interest rate, and a period, **Equation 3-4** is used:

$$F_n = A\left[\frac{(1+i)^n - 1}{i}\right] \tag{3-4}$$

3.1 Life Cycle Cost/Benefit Analysis

The life cycle cost/benefit analysis (LCCBA) is a comprehensive example of a second-level method. LCCBA computes a representative cost to allow a direct comparison of all systems under consideration. As with all second-level methods, the time value of money is considered. An outline of the LCCBA method is illustrated in **Table 3-1**. Here, two lighting system alternatives (System I and System II) are compared. A comparison of additional options, if desired, would follow the same format. The details of Sections A and B in the table are discussed in **Sections 3.1.2** and **3.1.3**, respectively.

Table 3-1. Example of a Life Cycle Cost/Benefit Analysis Method for Two Systems

Section A – Initial Costs	System I	System II
1. System, labor, and material (luminaire, incl. lamps, ballasts; controls)		
2. Total kw lighting		
3. Tons of air conditioning required to dissipate heat of lighting*		
4. First cost of air conditioning tons (line A.3) @ $____________/ton*		
5. Reduction in first cost of heating equipment		
6. Utility incentives		
7. Other first costs		
8. Subtotal mechanical and electrical installed cost		
9. Initial taxes		
10. Total costs		
11. Installed cost per square foot or per square meter**		
12. Watts of lighting per square foot or per square meter**		
13. Salvage value at end of economic life		

(continued on next page)

Table 3-1. Example of a Life Cycle Cost/Benefit Analysis Method for Two Systems *(continued)*

Section B – Annual Power and Maintenance Costs	System I	System II
1. Lighting utility costs = (burning hours)(kw)($/kwh)		
2. Air conditioning = (operating hours)(tons) (kw/ton)($/kwh)*		
3. Air conditioning maintenance = (tons)($/ton)*		
4. Reduction in heating cost		
5. Reduced heating maintenance = (Btu)($/Btu)		
6. Other annual costs		
7. Lamp replacement costs, if applicable		
8. Ballast replacement costs, if applicable		
9. Luminaire cleaning costs = (# of luminaires)($/luminaire)		
10. Annual insurance cost		
11. Annual property tax cost		
12. Income tax effect, *T*		
13. Total annual costs		

Table notes:

* The U.S. Customary System (USCS) unit "ton" refers to refrigeration: the time rate of cooling, equal to 12,000 British thermal units per hour (BTU/h, approximately 3,517 watts). If the required heat energy data is obtained in joules (J), it will be necessary to make a conversion to BTU before recording it (divide number of joules by 1,055 to obtain BTUs).

** These are side calculations that may be performed if the user desires. They do not affect the lighting calculations and comparisons.

3.1.1 General Considerations Applying the LCCBA. Since the analysis of lighting system economics is predominantly the analysis of costs, the convention used is that costs are positive values, and revenues, savings, and benefits are negative values. Estimates and default values (if mentioned) should only be used as a last resort when actual figures are not available. Some significant costs may include, but are not limited to: energy costs, maintenance costs, new construction costs needed for the new lighting system (e.g., ceiling and cove materials), materials costs, labor costs, and external costs. External costs may include, for example, meeting environmental standards, risk and liability costs for making a poor lighting decision, and avoided costs—the possible avoidance or downsizing of a significant purchase, such as a larger chiller or new power plant.

Once all the data for **Table 3-1** are collected, System I and System II may be compared either on an annual basis (cost per year of owning each system) or on a present-value basis.

3.1.2 Line-by-Line Analysis of Initial Costs. This section refers to Section A of **Table 3-1** (**Section 3.0**).

Line A.1. An estimate is prepared for the first cost of material and labor of the lighting system.

Line A.2. This is the total load (kilowatts, kW) of the lighting system.

Line A.3. Each lighting system introduces heat into the building, which should be dissipated by the air-conditioning system. One ton of air conditioning (cooling) can dissipate 12,000 Btu/hour of heat generated by the lighting system, or 3.517 kW of lighting load. Therefore, line A.2 is divided by 3.517kW/ton to obtain the value for line A.3.

Line A.4. Enter the first cost of the air-conditioning equipment of line A.3. First cost of machinery will vary from $1,000 to $2,000 per ton. Differently sized units can vary significantly in costs per ton of output capacity. It is important to keep in mind that if the cooling requirement of System I and System II only differ by a few tons, the same nominal size cooling plant may be used for each. For example, if System I requires 62 tons of cooling and System II requires 64, then each design might call for a 65-ton unit, and no first-cost differential would be realized. A similar caveat applies to line A.5. Line A.2 is multiplied by line A.3 to obtain the value for line A.4.

Line A.5. The heat generated by each lighting system may reduce the heating load on the building. This means that the heating plant may be smaller and the first cost of the heating plant reduced. The amount of that reduction for each lighting system is entered on line A.5. This will be a negative number. Heating equipment costs range from approximately $12 to $35 per million BTU, where one million BTUs = 293 kilowatt-hours (kWh). The same "per million BTU" cost is used for each alternative.

Important: The sizing of heating equipment would be based on when the maximum heat load occurs in a building. In most commercial buildings, the maximum heat load occurs when the building is warming up. *If the lights are off when the building is warming up, there is no contribution of the lighting system to reducing the maximum heat load*.

Line A.6. In order to reduce peak demand, some electric utility companies offer lighting rebate programs that provide incentives for end users who retrofit or install energy efficient lighting equipment in their buildings. Two types of programs are commonly available: technology-based and performance-based.

Technology-based utility rebate programs offer cash rebates for each energy-efficient component installed. These may include lamps, luminaires, and/or lighting controls. Performance-based programs offer a rebate for each kilowatt-hour (kWh) reduced in the total lighting energy load. In some areas, rebates are available for new construction as well as for retrofits. Whether a project is new or a retrofit, rebates vary depending on the utility company. Therefore, the lighting professional should consult the utility that serves the site under analysis. Any rebates are entered as a negative number.

Line A.7. Any other differential initial costs that are to be included (such as insulation, solar equipment, and tax credits) are entered here. Many older lighting systems may have lamps and ballasts that should be recycled, and disposal costs would be included here as a first cost.

Line A.8. The subtotal will be the sum of lines A.1, A.4, A.5, A.6, and A.7, taking proper note of signs.

Line A.9. For this line, the local sales tax rate is multiplied by the subtotal from line A.8.

Line A.10. This is the sum of lines A.8 and A.9.

Lines A.11 and A.12. These lines can be included as memos; they do not enter into the LCCBA calculation.

Line A.13. This is the amount the system will be worth at the end of its economic life (as scrap, for example). Use the same life for each system under comparison.

Note: This value is negative if money is received for the salvage; it is positive if a cost is incurred to dispose of the system at the end of its life. (See the comments in **Section 3.1.4** on environmental costs and in **Section 3.1.6** on present value calculations.)

3.1.3 Line-by-Line Analysis of Annual Power and Maintenance Costs. This section refers to Section B of **Table 3-1** (**Section 3.0**).

Line B.1. The number of burning hours and $/kWh will depend upon occupancy schedules and power rates. For example, a schedule of ten hours a day, five days a week, 52 weeks per year, represents 2,600 burning hours, which might be considered typical of a commercial building. Actual building use surveys should be conducted to obtain an accurate figure. Cleaning, weekend work, and flexible schedules can greatly increase actual annual hours of operation. Discrete lighting loggers can be used to measure "on time" as well as occupancy. In the U.S., the average energy cost of commercial, institutional, and industrial customers is approximately twelve cents per kWh, but one should use actual costs from the utility or from evaluating the user's electric bill. The figure used should include fuel charges, demand charges, and any other expense associated with energy use. A blended power rate can be effective if certain details about the actual rate are unavailable from the local electrical utility.

Control systems such as time of day controls, occupancy sensors, and daylight sensors can modify the burning hours significantly, affecting the annual power requirement of the lighting system being considered.

Line B.2. The tons of air conditioning should come from line A.3.

Line B.3. This value can be approximated by $165/ton times the air conditioning tons from line A.3.

Line B.4. This is the reduction in cost for fuel for heating equipment because of increased heat obtained from the lighting system. The fuel displaced by the lighting system operation only occurs when the lights are actually on. In most commercial buildings, (and this depends on the severity of the heating season), most heating BTUs are consumed during the warm up period. After people (who give off heat) occupy the building and as equipment is used, heat is added to the space. In some cases, the lighting has little impact on the amount of heat energy used.

$$\textit{Heating hours} = [(\textit{lighting hours})(0.85)] - [\textit{cooling hours}] \quad \textbf{(3-5)}$$

$$\textit{Number of ``Million BTUs''} = (\textit{kW of lighting})(\textit{heating hours}) / [293\ \mathrm{kWh/(megaBTU)}] \quad \textbf{(3-6)}$$

To convert costs of typical fuels to dollars per million BTUs, the cost is multiplied by the efficiency and divided by the appropriate units conversion factor. For example:

Electric heat: $[(1.0)(\$/\mathrm{kwh})]/(3.415) = \$/(\text{million BTU})$

Oil heat: $[(0.6)(\$/\mathrm{gal})]/(142) = \$/(\text{million BTU})$

Gas heat: $[(0.6)(\$/\mathrm{MCF})]/(1.0) = \$/(\text{million BTU})$

Line B.5. Heating maintenance costs can be approximated by $2 per million BTUs, but this may also vary greatly depending upon application and system type.

Line B.6. Luminaire cleaning and maintenance costs should be entered here, along with any other annual lighting system expenses not otherwise accounted for in Section B. (*Note:* Omit cleaning – see notes for line B.9.)

Line B.7. The cost of lamps per year will depend on the relamping strategy. If *spot relamping* is used, then:

$$\textit{lamp cost per year} = \frac{(\textit{cost, spot relamping one lamp})(\textit{no. lamps in system})}{(\textit{lamp life})/(\textit{annual burning hours})} \quad \textbf{(3-7)}$$

For *group relamping*, the value to enter is the average number of lamps replaced per year multiplied by the cost per lamp of group relamping.

Line B.8. The annualized ballast costs are calculated:

$$annualized\ ballast\ costs\ = \frac{(replacement\ cost,\ one\ ballast)(no.\ ballasts\ in\ system)}{(ballast\ life)/(annual\ burning\ hours)} \quad \textbf{(3-8)}$$

In many cases, system owners may choose to re-ballast using new technology and treat ballast costs as a capital expense. LED drivers and replaceable LED printed circuit boards may also be considered in field **B.8**.

Line B.9. This is the total cost of cleaning the luminaires.

Line B.10. This is typically approximately 1% to 1.5% of first cost.

Line B.11. This is typically approximately 0% to 9% of first cost.

Line B.12. The income tax effect is the amount by which depreciation reduces the system owner's income tax liability. Standard accounting refers to this feature as a *depreciation tax shield*. Assuming the asset is depreciated by the same amount each year (straight-line depreciation), the annual depreciation amount (D) is found from:

$$D\ =\ \frac{initial\ cost}{system\ life,\ in\ years}\ , \quad \textbf{(3-9)}$$

where *initial cost* is from line A.10.

If the system owner's income tax rate is t, expressed as a decimal fraction, then the tax effect, T, is given by:

$$T = (D)(t) \quad \textbf{(3-10)}$$

This tax effect is entered as a negative number, since it is a benefit or savings. Many times the tax laws will allow for an accelerated schedule of depreciation. The actual write-off life used by the system owner should always be employed.

Line B.13. This is the sum of lines B.1 through B.12.

3.1.4 Environmental Costs. Lighting equipment (not limited to lamps, ballasts, drivers, capacitors, controls, receptacles, reflectors, and refractors) may include materials that are hazardous or perceived hazardous when improperly disposed of at end of life. Some materials may be regulated by local, state, provincial, and/or federal environmental laws. End-of-life disposal costs may be significant and should be included in any complete economic analysis of lighting costs.

Choosing equipment with less regulated material, such as RoHS-compliant luminaires, or choosing equipment with longer life may reduce the future costs of lighting waste disposal.

3.1.5 Annual Cost Basis Conversions. Using **Equation 3-1** through **Equation 3-4** (see **Section 3.0**), the information from Section A and Section B of **Table 3-1** (**Section 3.0**) is converted to an equivalent annual cost. Since power and maintenance costs (line B.13) are already considered on an annual basis, they only need to be discounted into today's dollars. The first costs (line A.10) are converted to annual equivalents by rearranging **Equation 3-3**. Salvage values (line A.13) are converted to annual equivalents by rearranging **Equation 3-4**. Therefore, the total annual equivalent cost, *TAEC*, is computed for each system from:

$$TAEC\ =\ (\mathrm{B}\,.\,13\ value) + (\mathrm{A}\,.\,10\ value)\left[\frac{i(1+i)^n}{(1+\mathrm{i})^n - 1}\right] + (\mathrm{A}\,.\,13\ value)\left[\frac{i}{(1+i)^n - 1}\right]\ , \quad \textbf{(3-11)}$$

where:

n = the number of equal, sequential periods (future years) under consideration

i = interest rate

$B.13$, $A.10$, and $A.13$ *values* are cost data taken from corresponding line numbers in **Table 3-1**

Since the TAECs are costs in comparable terms, the system with the lower total annual equivalent cost should be selected, assuming equal performance over the same period.

3.1.6 Present Value Basis Conversions. The data from **Table 3-1** (see **Section 3.0**) may also be converted to present values. The resulting analysis is theoretically identical to the analysis based on annual cost and will always cause the same system to be selected as would be chosen under the annual cost mode. To compare equivalent costs in the present, the power and maintenance costs (line B.13) are translated to present values using **Equation 3-3** (see **Section 3.0**). Infrequent or irregular cash flows that can be quantified—such as programmed group relamping, and LED board and driver upgrades—may easily be added to the analysis in this present value (PV) conversion. The salvage values (line A.13) are discounted to the present using **Equation 3-1**. The total present value (TP) is then calculated for each system:

$$TP = (\mathrm{A.10}\ value)\ (\mathrm{B.13}\ value)\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] + (\mathrm{A.13}\ value)\left[\frac{i}{(1+i)^n}\right], \qquad \textbf{(3-12)}$$

where:

n = the number of equal, sequential time periods (future years) under consideration

i = interest rate

$A.10$, $B.13$, and $A.13$ *values* are cost data taken from corresponding line numbers in **Table 3-1**

Now the TPs from the two systems can be directly compared and the system with the lower cost chosen, again assuming both systems meet the design criteria.

3.2 Savings Investment Ratio

Another second-level method frequently encountered is the *savings investment ratio* (SIR). It is the ratio of the present value of future savings to the initial cost of the investment. It is particularly useful in retrofit situations or applications where an investment is made today for the specific purpose of saving costs (such as energy and maintenance) in the future. The SIR is based on similar data as collected for **Figure 4-1** (see **Section 4 Sensitivity Analysis**). Net future savings (i.e., future savings minus future costs) are discounted to the present using **Equations 3-2** through **3-4** (see **Section 3.0**). This value is compared with the initial investment, or the amount which would be spent today in order to generate these future savings. If the savings are greater than the cost (i.e., the SIR is greater than 1), then the investment may be profitable. However, the investment should not be made if the return percentage is less than the system owner's opportunity rate.

3.3 Notes on Interest Rates

A common question encountered in using methods that consider the time value of money is what interest rate to use. Using a low interest rate will make events in the future have a relatively greater impact on the decision. High interest rates, on the other hand, focus attention more on first costs; future events become less important.

Since the interest rate affects the outcome, it is important to give some consideration to making the proper choice. Most financial experts agree that the appropriate interest rate to use is the one that reflects the business owner's average cost of using other people's money—i.e., the owner's cost of capital. In general, businesses get their operating capital from two general sources: they sell stock (equity capital), and they use debt instruments such as bonds. The firm's cost of capital then is the weighted average of the cost of all its outstanding debt and equity instruments. This is found by taking the current market yield on each instrument (e.g., various classes of stock, all outstanding bond issues) and weighting that yield according to the total market value of the instrument. Thus, if a firm were financed 60% by equity yielding 10%, and 40% by debt yielding 8%, then the *weighted average cost of capital* would be: (0.6)(0.10) + (0.4)(0.08) = 0.092 = 9.2%.

Though the weighted average cost of capital is the appropriate interest rate to use, its use may not be justified for certain lighting analyses. For example, if the nature of the other data that enter the calculation is very approximate, calculating the cost of capital precisely may not be warranted. If this is the case, or if the lighting professional is not sufficiently familiar with the cost-of-capital concept, or if the data are not readily available, some other interest rate may be substituted, such as the prime rate currently charged by major lending institutions.

3.4 Internal Rate of Return

Some have suggested that the question of which interest rate to use can be avoided by assuming that the initial investment just equals the present worth of future savings, and calculating the implied discount rate. This is another second-level method known as the *internal rate of return* (IRR). In general, an iterative technique is required. An assumption is made about the interest rate. Then, using that rate, future values are discounted to the present using **Equations 3-1** through **3-4** (see **Section 3.0**). If the sum of the resulting present values does not equal the initial investment, the interest rate is adjusted and the process repeated. This is continued until an interest rate is found that equates investment with present worth of savings. That interest rate is the IRR. Most rigorous second-level spreadsheets can calculate a project's IRR instantaneously.

It can be seen, however, that this does not really alleviate the question of which interest rate to use. The IRR should still be compared with something in order to determine whether the project is acceptable. For example, assuming an IRR of 10% is computed for a certain system, would this be a sufficient return on the investment? The appropriate response is yes if 10% exceeds the firm's cost of capital. Obviously, the discussion has come full circle. It is important to note that the internal rate of return indicates *whether* a project is profitable, but not *how* profitable it is.

3.5 Net Present Value

One of the most widely recognized methods of financial analysis by individuals in the finance and investment industry is the *net present value* (NPV) method. When using capital budgeting techniques to evaluate which of several projects an organization needs to choose between, the financial team's most important consideration is often discovering which of these options will produce the greatest financial value to the firm, in terms of either reduction in operating expenses or improvement in another area of net income generation. In this manner, an NPV analysis can compare very different investment projects, such as a lighting upgrade that can reduce a firm's operating expenses and that is a low-risk investment, or acquisition of another company, which may be larger in scale and perhaps also in uncertainty.

Keeping in mind the importance of the time value of money (see **Section 3.0**), it is important to note that NPV is a second-level method that fully discounts all future income streams. Many companies use NPV as an analysis tool to make wise investment decisions; e.g., does the company spend "x" amount of money to open up a new product line that initially will be an enormous cash drain but in three years will be an excellent income producer, or does the company spend less money trying to revitalize its current product, recognizing that future growth potential will be less? NPV will help any firm make this kind of important decision. This same method will work equally well for examining cost savings options.

Equation 3-13 is used to find the net present value for a project:

$$NPV = \sum_{t=0}^{n}\left[\frac{CFAT_t}{(1+i)^t}\right] \quad , \tag{3-13}$$

where:

n = the number of equal, sequential periods (future years) under consideration

t = the time, which runs sequentially from 0 years to n years

$CFAT_t$ = the cash flow after taxes during future year t (when $t = 0$, then $CFAT$ = the initial investment)

i = the prevailing interest rate during future year t

Today's NPV is the initial investment (entered in the NPV calculation as a negative number) plus the sum of all "future year" calculations up to and including n, the last future year being considered. Therefore, today's NPV is calculated with $n = 0$, $n = 1$, $n = 2$, … , $n =$ last future year; then all these results are summed.

For example, if the initial investment is $10,000, $n = 4$, the CFAT for the first and second year is $4,000 each year, the CFAT for the third and fourth year is $3,000 each year, and the interest rate during all years is 10%, then entering this data into **Equation 3-5**, NPV yields:

$$NPV = -\$10{,}000 + \frac{\$4{,}000}{(1+0.1)^1} + \frac{\$4{,}000}{(1+0.1)^2} + \frac{\$3{,}000}{(1+0.1)^3} + \frac{\$3{,}000}{(1+0.1)^4}$$

$$\mathrm{NPV} = -\$10{,}000 + \$3{,}636.36 + \$3{,}305.79 + \$2{,}253.94 + \$2{,}049.04 = \$1{,}245.13$$

Discount tables are readily available once the cash flows by year have been estimated (see **Annex C** and **Annex D**.) Many electronic spreadsheet programs do factor in the time value of money, mainly through the use of the formulae already mentioned. If the spreadsheet is not that rigorous and ends up treating all cash flows the same (i.e., not discounting them), then the lighting practitioner should use the discount tables. All NPVs greater than zero should be considered, with the highest-NPV project normally favored. This method is especially effective in analyzing the economics of uneven cash flows from year to year; e.g., scheduled group relamping, and increased energy rates. Because the LCCBA, as a present-value metric, works in all cases, including new construction, remodel, and retrofit or upgrade projects, it is the method recommended in this document, even favored over NPV.

4.0 Sensitivity Analysis

One problem incurred with even a good model (such as the second-level methods) is that, once a model is selected as being the most appropriate, there is a temptation to adhere to the results as though they were chiseled in stone. Constant reminders are necessary that output from any model can be no better than the input. This leads directly to the suggestion that sensitivity analysis should be performed to determine the "stability" of the model's output for a given comparison.

In sensitivity analysis, questions are asked such as: How good are the assumptions serving as input into the model? Is the opportunity rate of interest really 8%? Or could it be 7%? Or 70%! What if ballasts are not replaced in the tenth year, but in the twelfth? Would it matter? Would such changes make a previously unacceptable alternative emerge as the most attractive?

Figure 4-1 illustrates a sensitivity analysis for the effect of various energy costs on two lighting systems, A and B. System A has a large, fixed maintenance component, but uses relatively little energy. System B requires little maintenance, but uses much more energy than A. Therefore, System B is more sensitive to energy costs than is System A. If one analysis is performed using, for example, 10 cents per kilowatt-hour ($0.10/kWh), then System B would appear to be less costly. If that single scenario were to choose $0.14/kWh as the energy cost, then the conclusion should be that System A would cost less to operate.

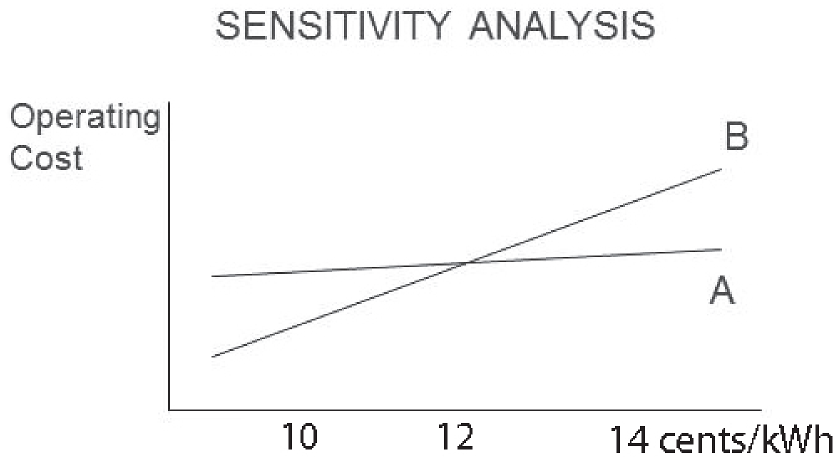

Figure 4-1. In this example, System B, which requires little maintenance but uses considerable energy, is much more sensitive to energy costs than System A, characterized by large fixed annual maintenance charges and relatively low energy consumption.

Similar analyses should be performed to determine whether the decision is sensitive to any of the other input variables for which the value or timing is uncertain. However, accountants warn that all costs and their timing, even those that have already occurred, are approximations! This invites consternation by indicating that rigorous economic analysis will require taking into account all of the uncertainties in all of the inputs simultaneously. There are many ways to approach this sticky problem. Indeed, volumes have been written on the subject of decision making under uncertainty.

A possible course for the lighting industry to pursue is likely to be one that continues to tap the power of the computer. For example, Monte Carlo simulations have been used to consider the effects of ranges of input values on the present worth of lighting systems.[2] Typically, the most sensitive variables have been the annual burning hours and cost per kilowatt-hour.

On the first pass the computer might randomly select an opportunity rate of 8.05%, ballast replacement in year 11, and an energy rate of $0.087/kWh as input values, and from these compute the present worth of the alternative. The second time through, it might randomly pick a 7.6% interest rate, ballast replacement in the tenth year, and $0.092/kWh as the energy rate, and again calculate the present worth. After numerous repetitions of such computations, a very clear picture emerges of the probability distribution of the present value of the alternative under consideration. This allows construction of confidence intervals. In the case of a 95% confidence interval, these are the present values *x* and *y* between which 95% of the sample present worth values fall. If the input is reliable, then it seems highly likely that the actual present worth of the project will also fall in that range. As most lighting economic analysis today is done on computer, it may be a consideration for the software writer to include a sensitivity analysis subprogram to check the results of all projects being evaluated.

5.0 Benefit Analysis

In most of the preceding discussion, quantification of the various costs that will occur over the life of a lighting system was the focus consideration. However, if only costs are considered, an important aspect of the situation—namely, benefits—is ignored. A lighting system is purchased for the benefits it produces. If a choice is made from among several alternative lighting systems based solely on cost, the benefit side of the equation has been ignored, or at best, some critical (often dubious) assumptions have been made. Usually, the implicit assumption is that all systems provide equal benefits, and therefore, the one that provides these benefits at the lowest cost is the "best."

However, an assumption of equal benefits is rarely true. Even if two systems are equivalent on one scale of benefits measurement (for example, they provide equal illuminance), it is likely that they will differ on others (such as glare, color rendering, or aesthetic appeal of the luminaire). If economic values could be assigned to these benefits, then the analysis would be straightforward; dollar amounts for benefits would be included in the present worth analysis in the same way that other revenues, such as salvage values, are included. However, many of the benefits of light are intangible and difficult to quantify. This is complicated by the fact that such benefits are intricately related; for example, the perceived pleasantness of a lighting system may involve a complex interaction of color, illuminance, and the activity being performed under the lighting.

For these reasons, the benefits of light are rarely (if ever) explicitly accounted for in lighting economics. Indeed, at the current level of knowledge, a research effort aimed at untangling all of the relationships among these issues would be difficult and highly subjective. It is, however in the purview of the lighting professional to evaluate various qualitative measures, through careful study of potential lighting systems using manufacturers' literature, samples, and/or mock ups, to ultimately write a specification for products that will ensure that the desired benefits are derived from the specified lighting system.

A radically different approach to the problem may also be considered. One proposed method would take advantage of the fact that people regularly make decisions about the benefits of light that have monetary consequences. The goal: making explicit that information which lighting designers and consultants already use implicitly in their decision-making process.

For example, in choosing between two lighting systems, a lighting engineer may select System A, even though it has a higher life-cycle cost than System B. The "non-quantifiable" factor leading to this choice may be that System A has superior color rendering properties. The owner might be willing to incur, say, a 10% greater cost for the benefit of better color. Would System A still have been chosen had System B been 20% less expensive? Or 30 percent? Through a series of such questions, a point of indifference could be found regarding the dollar value of color rendering.

Color is but one possible dimension that may enter into the decision-making process. In defining this quantification method, other dimensions will need to be determined. From there, a framework will be developed within which the designer can quickly determine weighting factors for various benefits on a particular project. However, the goal for this research would not be to assess specific benefits permanently, but rather to develop a methodology or framework that can assign fresh dollar values to the key dimensions of *any* project. Lighting professionals should keep abreast of developments on this topic so that they may provide their clients with a state-of-the-art economic analysis.

6.0 First-Level Analysis Methods

Second-level analysis methods have been shown to provide accurate information about the economic desirability of lighting projects. For reasons stated in **Section 2.2**, second-level methods are the only ones recommended by the IES without qualification, but there are several other widely used methods of economic analysis that may be considered. First-level analysis methods are discussed here so that the lighting professional can avoid the associated pitfalls and can respond intelligently to clients or other professionals who suggest relying upon such methods.

First-level analysis can be performed using simple methods, generally capable of being computed "by hand," i.e., without resorting to a computer. The minimal effort required for these techniques yields only coarse information that may be misleading. This is because these methods usually do not take into consideration the time value of money, tax effects, the duration of the product's life-cycle, or influences of the lighting system on the cost of other systems. Simple analysis may prove helpful in removing some alternatives from consideration, but when the decision is narrowed down to the final few alternatives, more-rigorous methods are required.

Another way of deciding whether or not to use a first-level method is to ask the question "what is the cost of a wrong answer?" If the price is minimal, then first-level methods can do little harm. If expenses can be significant, then a more thorough analysis is called for. This section provides a discussion of various first-level analysis methods. Again, caution needs to be exercised due to the potential for misleading results caused by the simplifying assumptions.

The most simplistic economic decision consists of one rule: compare initial cost and buy the cheapest product. For example, if lamp A costs $1.20 and lamp B costs $1.00, then lamp B would be purchased. If the lamps are identical in the service they provide, this may be a sufficient analysis. However, if lamp A produces 1,000 lumens but lamp B produces only 800 lumens, then the choice may be lamp A based on a cost-per-lumen comparison (A = 0.120 cents per lumen, while B = 0.125 cents per lumen). However, lamp A may only have a rated life of 1,000 hours as compared with 1,100 hours for lamp B, thus necessitating a further refinement of the calculation. These basic questions have been considered for years by many people in the lighting industry to help evaluate the economics of differing lighting products and to help sell their products to the public.

6.1 Simple Payback

One of the most common techniques used to obtain an answer is the *simple payback* method, which attempts to answer the question, how long will it take to recoup the initial investment in a lighting system? *Simple payback* is defined as the incremental initial cost of a system divided by the annual cash flow or savings that the system engenders:

$$simple\ payback = \frac{incremental\ investment}{incremental\ annual\ cash\ flow} \tag{6-1}$$

The term *incremental* in this definition indicates a comparison of one system to another. Two types of comparisons are possible: comparing a proposed replacement system to an existing system, or comparing design alternatives for a new space in which there is no "existing" system. The former is the more common usage of the simple payback concept in lighting. This method assumes a uniform series of cash flows, which may not be the case.

Frequently, a familiar question (appropriate to the payback method) arises concerning the replacement of existing lighting with new technology: how long will it take for the improvement to pay for itself? For example, the owner of an office building is considering replacing 1,000 standard fluorescent ballasts with a new energy efficient version. It is estimated that the new ballasts will save about $2,000 in energy costs per year. The expense associated with purchasing and installing the new ballasts is estimated at $11,000. Under these conditions the simple payback would be: \$11,000/\$2,000 = 5.5 years. (*Note:* These figures are totally fictitious and should in no way be used to justify an actual decision concerning ballast replacement).

Using the simple payback method to compare alternatives involves applying the same formula to differences in investments and cash flows. An example that considers four lighting systems is shown in **Table 6-1**; these are the Base System, Alternative A, Alternative B, and Alternative C. Here, column I shows the initial investments in thousands of dollars. Column II compares the Alternatives with the Base and gives the increase above the Base cost of each Alternative. Annual operating and maintenance costs are shown in column III, and column IV gives differences in annual cost between the Alternative and the Base. In column V, the payback is calculated as the result of column II data divided by column IV data.

In traditional use of the payback method, Alternative C is removed from consideration since it does not pay back, and Alternate B is preferred to Alternative A since it pays back more quickly (even though the initial cost is higher). Finally, either Alternative B or the Base is selected depending upon whether 2.86 years is considered a rapid enough payback. Simple payback is widely used in the lighting industry today because it is easy to understand and use. Closer inspection reveals that, like all first-level methods, it may have significant drawbacks. In the example shown in **Table 6-1**, Alternative B looks like the better option when compared to Alternative A: a 2.86-year payback vs. a 3.33-year payback, respectively. However, there is no information here about cash flows over either system's life. For example, what happens to system economics in years 3, 4, 5 …? What is the life of each system? How long will the owner occupy the space? There are many kinds of alternative lighting systems that vary greatly in initial investment and 20-year life-cycle costs (and returns), in addition to the short-term returns revealed by the simple payback method.

A common error is to look at simple payback alone and then decide, for example, to make a simple lamp change that reduces energy consumption from 75 to 45 watts when a second alternative with a longer payback but greater system life exists that can also cut energy consumption from 75 watts to 15 watts. A more rigorous second-level analysis method would most likely recommend the more costly second alternative.

Table 6-1. Simple Payback Used to Compare Lighting System Designs

Systems	I. Initial Investments (k$)	II. Incremental Change from Base (k$)	III. Additional Annualized Costs (k$*)	IV. Annual Savings (k$*)	V. Simple Payback
Base System	110	n/a	20	n/a	n/a
Alternative A	120	10	17	3	10/3 = 3.33
Alternative B	130	20	13	7	20/7 = 2.86
Alternative C	140	30	21	-1	Does not pay back

*Relative to the Base System case

Simple payback *may* be suitable as a "Go/No Go" test for *one* option. For example: A company permits any investment in its facilities to be implemented that has a three-year (or shorter) payback. However, if a project that has a simple payback of better than the desired rate is accepted for implementation, how does the designer know whether the *best choice* has been made? Too often, the designer runs the simple payback analysis on alternative projects and then picks the project with the quickest payback. This is when a second-level method is needed to compare differing lighting options.

Simple payback should not be used as a determinant between two or more acceptable lighting system alternatives. In fact, it is actually a risk assessment tool posing as a profitability metric. This is seen from analysis of the question the method seeks to answer. It does not answer the question "is a certain investment profitable?" Rather, it assuages the concerns of someone unsure about the future and anxious to recoup an investment as soon as possible. However, if getting money back is the primary concern, why invest at all? If no investment is made, then that money is available immediately and the payback is zero years, the ideal result of a simple payback calculation! The bottom line is, using the simple payback method will not lead to an unprofitable investment being chosen, but it may not lead to choosing the *most* profitable investment.

Another problem with simple payback is that it fails to consider what happens *after* the investment is repaid. For example, if Alternative A in **Table 6-1** continues to save money for ten years, but the savings of Alternative B decline sharply each year after the payback period, the simple payback method will not capture this and will suggest selection of the inferior alternative. A similar problem would be encountered if the alternative systems have different lengths of economic life. Therefore, the simple payback method should not be used when the alternatives have non-uniform cash flows. It is inappropriate for options having different useful lives, and should be used with skepticism even in cases of equivalent lifetime.

6.2 Simple Rate of Return

The simple rate of return is just the inverse of simple payback (see **Equation 6-1**, **Section 6.1**):

$$\textit{simple rate of return} = \frac{\textit{incremental annual cash flow}}{\textit{incremental investment}} \tag{6-2}$$

Thus, if a system saves \$20,000 per year and requires an incremental (initial) investment of \$100,000, then its simple return is \$20,000/\$100,000 or 20% (or a five-year simple payback). The advantages and disadvantages are identical to the simple payback method: Simple rate of return is easy to apply and understand, but it cannot deal with non-uniform saving streams or options with dissimilar lifetimes. Both simple payback and simple rate of return ignore the cost of capital and interest rates.

6.3 Cost of Light

An older method of analysis that has specific use when only two lamp options are being evaluated is called *cost of light*. Historically, it addressed the issue of A-type incandescent lamps compared to other incandescent lamps with a longer life but lower lumen output.[3] The cost-of-light model basically considers the lumen output of the two sources measured in mean lamp lumens against the owning and operating costs expended to operate each lamp type. **Equation 6-3** provides a means of considering the performance of a given light source by permitting calculation of the dollar cost per million lumen-hours (U) for each lamp type:

$$U = \left[\frac{(F+M)}{H} + (R)(W) + \frac{(P+h)}{L}\right]\left[\frac{10}{Q}\right], \tag{6-3}$$

where:

U = cost in dollars per million lumen-hours

F = purchase cost (in cents per luminaire per year)

M = cleaning cost (in cents per luminaire per year)

H = burning hours per year (in thousands)

R = energy rate (in cents per kWh)

W = input watts per luminaire

P = lamp cost (in cents) per luminaire

h = spot relamp cost (in cents) per luminaire

L = lamp life (in thousands of hours)

Q = mean lamp lumens

When comparing several lamp types, the one with the lowest cost of light would have the best economic value to the user. Some manufacturers have guides that compare various lamp types using a cost-of-light analysis. However, because the formula does not take into consideration the luminaire, lamp, and ballast losses, or various depreciation factors (such as dirt), it is limited to making lamp-to-lamp economic comparisons among similar lamp types.

The cost-of-light method of economic analysis is included in this discussion of first-level techniques because in its application, annual owning and operating expenses and future cash flows are discounted.

Annex A – Notes on Financial Analysis

Note: This Annex is not part of ANSI/IES RP-31-20, Recommended Practice: Economic Analysis of Lighting. *It is provided for informational purposes only.*

There are several different ways to consider some of the elements in a financial analysis of a lighting system. The most significant categories are:

- Initial costs
- Operating costs
- Financial incentives and utility rebates
- Time of operation and reduction of time through the use of lighting controls
- Thermal gains and losses (HVAC considerations)
- The costs of alternative energy sources
- Electric energy costs offset through the use of daylighting

Many times lighting systems are not truly equal in performance. Luminaires may have different wattages, efficiencies, lumen outputs, lumen maintenance factors, and lifetimes. It is an important consideration for the lighting professional to attempt to make various systems equal in performance. One common technique is to start by making their average maintained illuminances equal by generating a factor to equate luminous flux between all alternative luminaires. One can equate illuminances at a site by creating a multiplier for each system to allow it to be compared to the base system. If the base system is 4,000 lumens, average maintained, and System A is 4,400 lumens, it follows that fewer luminaires of the alternative than the base system would be needed. In this case, one could use a multiplier of 0.91 (4,000/4,400) with the actual luminaire count to equate the two alternatives. This is a necessary part of any financial analysis. If a design with 150 luminaires of the base 4,000-lumen type was acceptable, the alternative design with the 4,400-lumen type would only require (150)(0.91) = 136 luminaires.

In many scenarios, the designer does not have the flexibility to change luminaire placement, such as in a small room with a grid ceiling. The 9% material reduction of the previous example would not be practical in an area with only six luminaires spaced out in two rows of three. In that case, all alternatives should be very close to the original lumen package in order to be considered. There is no economic advantage in over-lighting a space. In all cases, performance of the alternative products can and should be considered. Cost reductions can be realized by reducing the number of luminaires in a project because of superior performance, but there are obvious limits to using an inappropriate product just because it has triple the raw lumens. Light distribution and glare are also important considerations. Ultimately, it is the design professional's decision about which luminaire and source type to specify or select. The financial analysis is merely a tool to help in rationalizing a specific alternative based upon equal performance.

Solid-state luminaires are generally being designed to replace one-for-one their fluorescent and HID counterparts that have in the past been de facto building standards. However, it is important for the designer to carefully consider the delivered lumens of each product considered. For example, for the lighting in a surgical operating room, utilizing a troffer with three times the lumens but requiring only four troffers total may not be as effective at meeting the lighting requirements as utilizing ten luminaires with a "normal" lumen package. Lighting software should be used to ensure that all spaces are lighted appropriately and meet IES recommendations for the particular application.

Annex B – Lighting Controls and Annual Burning Hours

Note: This Annex is not part of ANSI/IES RP-31-20, Recommended Practice: Economic Analysis of Lighting. *It is provided for informational purposes only.*

A current emphasis for professionals involved in energy management and electric load reduction is the increased use of lighting controls. LED luminaire technology, in particular, is suitable to incorporating integrated and add-on lighting controls without any appreciable loss in efficacy or source life. Not only are controls required by ordinance, but they are a highly effective way to reduce illumination when appropriate and thus reduce energy use. Programmed lumen management, which saves additional energy throughout the early life of the lighting system, daylight sensing, and occupancy sensing are very practical methods of reducing owning and operating costs due to lighting.

One input to the financial analysis that is seldom changed is the field entitled *annual hours of operation*. It is a simple matter to modify this number to reflect the use of controls for a given lighting system. Adding controls to account for occupancy, daylighting, automatic reduction of some of the lighting based upon a preprogrammed time of day, or other control schemes, can be accounted for by reducing the hours of operation for those systems utilizing lighting controls.

Annex C – Present Value Tables

Note: This Annex is not part of ANSI/IES RP-31-20, Recommended Practice: Economic Analysis of Lighting. *It is provided for informational purposes only.*

Present value tables make it easy to discount any future event into today's dollars. As an example, if a group relamp costing an estimated $8,500.00 for lamps and labor was planned for year six, at an employer's cost of capital of 4%, that singular expense would be accounted for as $8,500.00 x 0.7903 = $6,717.55 in today's dollars. (See **Table C-1a**, Period 6, 4% column.) The present value would be added to the LCCBA formula as a year-six expenditure.

Note: With a good financial analysis program, the use of these tables may not be necessary. To calculate all future cash flows by hand using the formulas in this document, however, would be quite tedious, although it could be done.

Table C-1a. Present Value Table

Periods	1%	2%	3%	4%	5%	6%	7%
1	.9901	.9804	.9709	.9615	.9524	.9434	.9346
2	.9803	.9612	.9426	.9246	.9070	.8900	.8734
3	.9707	.9423	.9151	.8890	.8638	.8396	.8163
4	.9610	.9238	.8885	.8548	.8227	.7921	.7629
5	.9515	.9057	.8626	.8219	.7835	.7473	.7130
6	.9420	.8880	.8375	.7903	.7462	.7050	.6663
7	.9327	.8706	.8131	.7599	.7107	.6651	.6228
8	.9235	.8535	.7894	.7307	.6768	.6274	.5820
9	.9143	.8368	.7664	.7026	.6446	.5919	.5439
10	.9053	.8203	.7441	.6756	.6139	.5584	.5083
11	.8963	.8043	.7224	.6496	.5847	.5268	.4751
12	.8874	.7885	.7014	.6246	.5568	.4970	.4440
13	.8787	.7730	.6810	.6006	.5303	.4688	.4150
14	.8700	.7579	.6611	.5775	.5051	.4423	.3878
15	.8613	.7430	.6419	.5553	.4810	.4173	.3624
16	.8528	.7284	.6232	.5339	.4581	.3936	.3387
17	.8444	.7142	.6050	.5134	.4363	.3714	.3166
18	.8360	.7002	.5874	.4936	.4155	.3503	.2959
19	.8277	.6864	.5703	.4746	.3957	.3305	.2765
20	.8195	.6730	.5537	.4564	.3769	.3118	.2584
21	.8114	.6598	.5375	.4388	.3589	.2942	.2415
22	.8034	.6468	.5219	.4220	.3419	.2775	.2257
23	.7954	.6342	.5067	.4057	.3256	.2618	.2109
24	.7876	.6217	.4919	.3901	.3101	.2470	.1971
25	.7798	.6095	.4776	.3751	.2953	.2330	.1842
26	.7720	.5976	.4637	.3607	.2812	.2198	.1722
27	.7644	.5859	.4502	.3468	.2678	.2074	.1609
28	.7568	.5744	.4371	.3335	.2551	.1956	.1504
29	.7493	.5631	.4243	.3207	.2429	.1846	.1406
30	.7419	.5521	.4120	.3083	.2314	.1741	.1314
31	.7346	.5412	.4000	.2965	.2204	.1643	.1228
32	.7273	.5306	.3883	.2851	.2099	.1550	.1147
33	.7201	.5202	.3770	.2741	.1999	.1462	.1072
34	.7130	.5100	.3660	.2636	.1904	.1379	.1002
35	.7059	.5000	.3554	.2534	.1813	.1301	.0937
36	.6989	.4902	.3450	.2437	.1727	.1227	.0875
37	.6920	.4806	.3350	.2343	.1644	.1158	.0818
38	.6858	.4712	.3252	.2253	.1566	.1092	.0765
39	.6784	.4619	.3158	.2166	.1491	.1031	.0715
40	.6717	.4529	.3066	.2083	.1420	.0972	.0668
41	.6650	.4440	.2976	.2003	.1353	.0917	.0624
42	.6584	.4353	.2890	.1926	.1288	.0865	.0583
43	.6520	.4268	.2805	.1852	.1227	.0816	.0545
44	.6454	.4184	.2724	.1780	.1169	.0770	.0509
45	.6391	.4102	.2644	.1712	.1113	.0727	.0476
46	.6327	.4022	.2567	.1646	.1060	.0685	.0445
47	.6265	.3943	.2493	.1583	.1009	.0647	.0416
48	.6203	.3865	.2420	.1522	.0961	.0610	.0389
49	.6141	.3790	.2350	.1463	.0916	.0575	.0363
50	.6080	.3715	.2281	.1407	.0872	.0543	.0339

Table C-1b. Present Value Table

Periods	8%	9%	10%	11%	12%	13%	14%
1	.9259	.9174	.9091	.9009	.8929	.8850	.8772
2	.8573	.8417	.8264	.8116	.7972	.7831	.7695
3	.7938	.7722	.7513	.7312	.7118	.6931	.6750
4	.7350	.7084	.6830	.6587	.6355	.6133	.5921
5	.6806	.6499	.6209	.5935	.5674	.5428	.5194
6	.6302	.5963	.5645	.5346	.5066	.4803	.4556
7	.5835	.5470	.5132	.4817	.4523	.4251	.3996
8	.5403	.5019	.4665	.4339	.4039	.3762	.3506
9	.5002	.4604	.4241	.3909	.3606	.3329	.3075
10	.4632	.4224	.3855	.3522	.3220	.2946	.2697
11	.4289	.3875	.3505	.3173	.2875	.2607	.2366
12	.3971	.3555	.3186	.2858	.2567	.2307	.2076
13	.3677	.3262	.2897	.2575	.2292	.2042	.1821
14	.3405	.2992	.2633	.2320	.2046	.1807	.1597
15	.3152	.2745	.2394	.2090	.1827	.1599	.1401
16	.2919	.2519	.2176	.1883	.1631	.1415	.1229
17	.2703	.2311	.1978	.1696	.1456	.1252	.1078
18	.2502	.2120	.1799	.1528	.1300	.1108	.0946
19	.2317	.1945	.1635	.1377	.1161	.0981	.0829
20	.2145	.1784	.1486	.1240	.1037	.0868	.0728
21	.1987	.1637	.1351	.1117	.0926	.0768	.0638
22	.1839	.1502	.1228	.1007	.0826	.0680	.0560
23	.1703	.1378	.1117	.0907	.0738	.0601	.0491
24	.1577	.1264	.1015	.0817	.0659	.0532	.0431
25	.1460	.1160	.0923	.0736	.0588	.0471	.0378
26	.1352	.1064	.0839	.0663	.0525	.0417	.0331
27	.1252	.0976	.0763	.0597	.0469	.0369	.0291
28	.1159	.0895	.0693	.0538	.0419	.0326	.0255
29	.1073	.0822	.0630	.0485	.0374	.0289	.0224
30	.0994	.0754	.0573	.0437	.0334	.0256	.0196
31	.0920	.0691	.0521	.0394	.0298	.0226	.0172
32	.0852	.0634	.0474	.0355	.0266	.0200	.0151
33	.0789	.0582	.0431	.0319	.0238	.0177	.0132
34	.0730	.0534	.0391	.0288	.0212	.0157	.0116
35	.0676	.0490	.0356	.0259	.0189	.0139	.0102
36	.0626	.0449	.0323	.0234	.0169	.0123	.0089
37	.0580	.0412	.0294	.0210	.0151	.0109	.0078
38	.0537	.0378	.0267	.0190	.0135	.0096	.0069
39	.0497	.0347	.0243	.0171	.0120	.0085	.0060
40	.0460	.0318	.0221	.0154	.0107	.0075	.0053
41	.0426	.0292	.0201	.0139	.0096	.0067	.0046
42	.0395	.0268	.0183	.0125	.0086	.0059	.0041
43	.0365	.0246	.0166	.0112	.0076	.0052	.0036
44	.0338	.0226	.0151	.0101	.0068	.0046	.0031
45	.0313	.0207	.0137	.0091	.0061	.0041	.0028
46	.0290	.0190	.0125	.0082	.0054	.0036	.0024
47	.0269	.0174	.0113	.0074	.0059	.0032	.0021
48	.0249	.0160	.0103	.0067	.0043	.0028	.0019
49	.0230	.0147	.0094	.0060	.0039	.0025	.0016
50	.0213	.0134	.0085	.0054	.0035	.0022	.0014

Table C-1c. Present Value Table

Periods	15%	16%	17%	18%	19%
1	.8696	.8621	.8547	.8475	.8403
2	.7561	.7432	.7305	.7182	.7062
3	.6575	.6407	.6244	.6086	.5934
4	.5718	.5523	.5337	.5158	.4987
5	.4972	.4761	.4561	.4371	.4190
6	.4323	.4104	.3898	.3704	.3521
7	.3759	.3538	.3332	.3139	.2959
8	.3269	.3050	.2848	.2660	.2487
9	.2843	.2630	.2434	.2255	.2090
10	.2472	.2267	.2080	.1911	.1756
11	.2149	.1954	.1778	.1619	.1476
12	.1869	.1685	.1520	.1372	.1240
13	.1625	.1452	.1299	.1163	.1042
14	.1413	.1252	.1110	.0985	.0876
15	.1229	.1079	.0949	.0835	.0736
16	.1069	.0930	.0811	.0708	.0618
17	.0929	.0802	.0693	.0600	.0520
18	.0808	.0691	.0592	.0508	.0437
19	.0703	.0596	.0506	.0431	.0367
20	.0611	.0514	.0433	.0365	.0308
21	.0531	.0443	.0370	.0309	.0259
22	.0462	.0382	.0316	.0262	.0218
23	.0402	.0329	.0270	.0222	.0183
24	.0349	.0284	.0231	.0188	.0154
25	.0304	.0245	.0197	.0160	.0129
26	.0264	.0211	.0169	.0135	.0109
27	.0230	.0182	.0144	.0115	.0091
28	.0200	.0157	.0123	.0097	.0077
29	.0174	.0135	.0105	.0082	.0064
30	.0151	.0116	.0090	.0070	.0054
31	.0131	.0100	.0077	.0059	.0046
32	.0114	.0087	.0066	.0050	.0038
33	.0099	.0075	.0056	.0042	.0032
34	.0086	.0064	.0048	.0036	.0027
35	.0075	.0055	.0041	.0030	.0023
36	.0065	.0048	.0035	.0026	.0019
37	.0057	.0041	.0030	.0022	.0016
38	.0049	.0036	.0026	.0019	.0013
39	.0043	.0031	.0022	.0016	.0011
40	.0037	.0026	.0019	.0013	.0010
41	.0032	.0023	.0016	.0011	.0008
42	.0028	.0020	.0014	.0010	.0007
43	.0025	.0017	.0012	.0008	.0006
44	.0021	.0015	.0010	.0007	.0005
45	.0019	.0013	.0009	.0006	.0004
46	.0016	.0011	.0007	.0005	.0003
47	.0014	.0009	.0006	.0004	.0003
48	.0012	.0008	.0005	.0004	.0002
49	.0011	.0007	.0005	.0003	.0002
50	.0009	.0006	.0004	.0003	.0002

Annex D – Present Value of an Annuity

Note: This Annex is not part of ANSI/IES RP-31-20, Recommended Practice: Economic Analysis of Lighting. *It is provided for informational purposes only.*

The *present value of an annuity* can be used if the annual cash flows, called *annuities*, are equivalent each year. If there were an escalation in costs over the succeeding years (such as with rising utility costs), this table would not be useful. However, if energy rates were assumed to be stable over the next number of periods (years), this table could be used. For example, if System A shows an energy reduction of $7,000.00 per year for the next ten years, at a 4% interest rate, compared to the Base System, all ten years could be discounted: $7,500.00 x 8.1109 = $60,831.75. (See **Table D-1a**, Period 10, 4% column.)

Note: With a good financial analysis program, the use of these tables may not be necessary. To calculate all future cash flows by hand using the formulas in this document, however, would be quite tedious, although it could be done.

Table D-1a. Present Value of Annuity Factors

Periods	1%	2%	3%	4%	5%	6%	7%
1	.9901	.9804	.9709	.9615	.9524	.9434	.9346
2	1.9704	1.9416	1.9135	1.8861	1.8594	1.8334	1.8080
3	2.9410	2.8839	2.8286	2.7751	2.7232	2.6730	2.6243
4	3.9020	3.8077	3.7171	3.6299	3.5460	3.4651	3.3872
5	4.8534	4.7135	4.5797	4.4518	4.3295	4.2124	4.1002
6	5.7955	5.6014	5.4172	5.2421	5.0757	4.9173	4.7665
7	6.7282	6.4720	6.2303	6.0021	5.7864	5.5824	5.3893
8	7.6517	7.3255	7.0197	6.7327	6.4632	6.2098	5.9713
9	8.5660	8.1622	7.7861	7.4353	7.1078	6.8017	6.5152
10	9.4713	8.9826	8.5302	8.1109	7.7217	7.3601	7.0236
11	10.3676	9.7868	9.2526	8.7605	8.3064	7.8869	7.4987
12	11.2551	10.5753	9.9540	9.3851	8.8633	8.3838	7.9427
13	12.1337	11.3484	10.6350	9.9856	9.3936	8.8527	8.3577
14	13.0037	12.1062	11.2961	10.5631	9.8986	9.2950	8.7455
15	13.8651	12.8493	11.9379	11.1184	10.3797	9.7122	9.1079
16	14.7179	13.5777	12.5611	11.6523	10.8378	10.1059	9.4466
17	15.5623	14.2919	13.1661	12.1657	11.2741	10.4773	9.7632
18	16.3983	14.9920	13.7535	12.6593	11.6896	10.8276	10.0591
19	17.2260	15.6785	14.3238	13.1339	12.0853	11.1581	10.3356
20	18.0456	16.3514	14.8775	13.5903	12.4622	11.4699	10.5940
21	18.8570	17.0112	15.4150	14.0292	12.8212	11.7641	10.8355
22	19.6604	17.6580	15.9369	14.4511	13.1630	12.0416	11.0612
23	20.4558	18.2922	16.4436	14.8568	13.4886	12.3034	11.2722
24	21.2434	18.9139	16.9355	15.2470	13.7986	12.5504	11.4693
25	22.0232	19.5235	17.4131	15.6221	14.0939	12.7834	11.6536
26	22.7952	20.1210	17.8768	15.9828	14.3752	13.0032	11.8258
27	23.5596	20.7069	18.3270	16.3296	14.6430	13.2105	11.9867
28	24.3164	21.2813	18.7641	16.6631	14.8981	13.4062	12.1371
29	25.0658	21.8444	19.1885	16.9837	15.1411	13.5907	12.2777
30	25.8077	22.3965	19.6004	17.2920	15.3725	13.7648	12.4090
31	26.5423	22.9377	20.0004	17.5885	15.5928	13.9291	12.5318
32	27.2696	23.4683	20.3888	17.8736	15.8027	14.0840	12.6466
33	27.9897	23.9886	20.7658	18.1476	16.0025	14.2302	12.7538
34	28.7027	24.4986	21.1318	18.4112	16.1929	14.3681	12.8540
35	29.4086	24.9986	21.4872	18.6646	16.3742	14.4982	12.9477
36	30.1075	25.4888	21.8323	18.9083	16.5469	14.6210	13.0352
37	30.7995	25.9695	22.1672	19.1426	16.7113	14.7368	13.1170
38	31.4847	26.4406	22.4925	19.3679	16.8679	14.8460	13.1935
39	32.1630	26.9026	22.8082	19.5845	17.0170	14.9491	13.2649
40	32.8347	27.3555	23.1148	19.7928	17.1591	15.0463	13.3317
41	33.4997	27.7995	23.4124	19.9931	17.2944	15.1380	13.3941
42	34.1581	28.2348	23.7014	20.1856	17.4232	15.2245	13.4524
43	34.8100	28.6616	23.9819	20.3708	17.5459	15.3062	13.5070
44	35.4555	29.0800	24.2543	20.5488	17.6628	15.3832	13.5579
45	36.0945	29.4902	24.5187	20.7200	17.7741	15.4558	13.6055
46	36.7272	29.8923	24.7754	20.8847	17.8801	15.5244	13.6500
47	37.3537	30.2866	25.0247	21.0429	17.9810	15.5890	13.6916
48	37.9740	30.6731	25.2667	21.1951	18.0772	15.6500	13.7305
49	38.5881	31.0521	25.5017	21.3415	18.1687	15.7076	13.7668
50	39.1961	31.4236	25.7298	21.4822	18.2559	15.7619	13.8007

Table D-1b. Present Value of Annuity Factors

Periods	8%	9%	10%	11%	12%	13%	14%
1	.9259	.9174	.9091	.9009	.8929	.8850	.8772
2	1.7833	1.7591	1.7355	1.7125	1.6901	1.6681	1.6467
3	2.5771	2.5313	2.4869	2.4437	2.4018	2.3612	2.3216
4	3.3121	3.2397	3.1699	3.1024	3.0373	2.9745	2.9137
5	3.9927	3.8897	3.7908	3.6959	3.6048	3.5172	3.4331
6	4.6229	4.4859	4.3553	4.2305	4.1114	3.9976	3.8887
7	5.2064	5.0330	4.8684	4.7122	4.5638	4.4226	4.2883
8	5.7466	5.5348	5.3349	5.1461	4.9676	4.7988	4.6389
9	6.2469	5.9952	5.7590	5.5370	5.3283	5.1317	4.9464
10	6.7101	6.4177	6.1446	5.8892	5.6502	5.4262	5.2161
11	7.1390	6.8052	6.4951	6.2065	5.9377	5.6869	5.4527
12	7.5361	7.1607	6.8137	6.4924	6.1944	5.9176	5.6603
13	7.9038	7.4869	7.1034	6.7499	6.4235	6.1218	5.8424
14	8.2442	7.7862	7.3667	6.9819	6.6282	6.3025	6.0021
15	8.5595	8.0607	7.6061	7.1909	6.8109	6.4624	6.1422
16	8.8514	8.3126	7.8237	7.3792	6.9740	6.6039	6.2651
17	9.1216	8.5436	8.0216	7.5488	7.1196	6.7291	6.3729
18	9.3719	8.7556	8.2014	7.7016	7.2497	6.8399	6.4674
19	9.6036	8.9501	8.3649	7.8393	7.3658	6.9380	6.5504
20	9.8181	9.1285	8.5136	7.9633	7.4694	7.0248	6.6231
21	10.0168	9.2922	8.6487	8.0751	7.5620	7.1016	6.6870
22	10.2007	9.4424	8.7715	8.1757	7.6446	7.1695	6.7429
23	10.3711	9.5802	8.8832	8.2664	7.7184	7.2297	6.7921
24	10.5288	9.7066	8.9847	8.3481	7.7843	7.2829	6.8351
25	10.6748	9.8226	8.0770	8.4217	7.8431	7.3300	6.8729
26	10.8100	9.9290	9.1609	8.4881	7.8957	7.3717	6.9061
27	10.9352	10.0266	9.2372	8.5478	7.9426	7.4086	6.9352
28	11.0511	10.1161	9.3066	8.6016	7.9844	7.4412	6.9607
29	11.1584	10.1983	9.3696	8.6501	8.0218	7.4701	6.9830
30	11.2578	10.2737	9.4269	8.6938	8.0552	7.4957	7.0027
31	11.3498	10.3428	9.4790	8.7331	8.0850	7.5183	7.0199
32	11.4350	10.4062	9.5264	8.7686	8.1116	7.5383	7.0350
33	11.5139	10.4644	9.5694	8.8005	8.1354	7.5560	7.0482
34	11.5869	10.5178	9.6086	8.8293	8.1566	7.5717	7.0599
35	11.6546	10.5668	9.6442	8.8552	8.1755	7.5856	7.0700
36	11.7172	10.6118	9.6765	8.8786	8.1924	7.5979	7.0790
37	11.7752	10.6530	9.7059	8.8996	8.2075	7.6087	7.0868
38	11.8289	10.6908	9.7327	8.9186	8.2210	7.6183	7.0937
39	11.8786	10.7255	9.7570	8.9357	8.2330	7.6268	7.0997
40	11.9246	10.7574	9.7791	8.9511	8.2438	7.6344	7.1050
41	11.9672	10.7866	9.7991	8.9649	8.2534	7.6410	7.1097
42	12.0067	10.8134	9.8174	8.9774	8.2619	7.6469	7.1138
43	12.0432	10.8380	9.8340	8.9886	8.2696	7.6522	7.1173
44	12.0771	10.8605	9.8491	8.9988	8.2764	7.6568	7.1205
45	12.1084	10.8812	9.8628	9.0079	8.2825	7.6609	7.1232
46	12.1374	10.9002	9.8753	9.0161	8.2880	7.6645	7.1256
47	12.1643	10.9176	9.8&66	9.0235	8.2928	7.6677	7.1277
48	12.1891	10.9336	9.8969	9.0302	8.2972	7.6705	7.1296
49	12.2122	10.9482	9.9063	9.0362	8.3010	7.6730	7.1312
50	12.2335	10.9617	9.9148	9.0417	8.3045	7.6752	7.1327

Table D-1c. Present Value of Annuity Factors

Periods	15%	16%	17%	18%	19%
1	.8696	.8621	.8547	.8475	.8403
2	1.6257	1.6052	1.5852	1.5656	1.5465
3	2.2832	2.2459	2.2096	2.1743	2.1399
4	2.8550	2.7982	2.7432	2.6901	2.6386
5	3.3522	3.2743	3.1993	3.1272	3.0576
6	3.7845	3.6847	3.5892	3.4976	3.4098
7	4.1604	4.0386	3.9224	3.8115	3.7057
8	4.4873	4.3436	4.2072	4.0776	3.9544
9	4.7716	4.6065	4.4506	4.3030	4.1633
10	5.0188	4.8332	4.6586	4.4941	4.3389
11	5.2337	5.0286	4.8364	4.6560	4.4865
12	5.4206	5.1971	4.9884	4.7932	4.6105
13	5.5831	5.3423	5.1183	4.9095	4.7147
14	5.7245	5.4675	5.2293	5.0081	4.8023
15	5.8474	5.5755	5.3242	5.0916	4.8759
16	5.9542	5.6685	5.4053	5.1624	4.9377
17	6.0472	5.7487	5.4746	5.2223	4.9897
18	6.1280	5.8178	5.5339	5.2732	5.0333
19	6.1982	5.8775	5.5845	5.3162	5.0700
20	6.2593	5.9288	5.6278	5.3527	5.1009
21	6.3125	5.9731	5.6648	5.3837	5.1268
22	6.3587	6.0113	5.6964	5.4099	5.1486
23	6.3988	6.0442	5.7234	5.4321	5.1668
24	6.4338	6.0726	5.7465	5.4509	5.1822
25	6.4641	6.0971	5.7662	5.4669	5.1951
26	6.4906	6.1182	5.7831	5.4804	5.2060
27	6.5135	6.1364	5.7975	5.4919	5.2151
28	6.5335	6.1520	5.8099	5.5016	5.2228
29	6.5509	6.1656	5.8204	5.5098	5.2292
30	6.5660	6.1772	5.8294	5.5168	5.2347
31	6.5791	6.1872	5.8371	5.5227	5.2392
32	6.5905	6.1959	5.8437	5.5277	5.2430
33	6.6005	6.2034	5.8493	5.5320	5.2462
34	6.6091	6.2098	5.8541	5.5356	5.2489
35	6.6166	6.2153	5.8582	5.5386	5.2512
36	6.6231	6.2201	5.8617	5.5412	5.2531
37	6.6288	6.2242	5.8647	5.5434	5.2547
38	6.6338	6.2278	5.8673	5.5452	5.2561
39	6.6380	6.2309	5.8695	5.5468	5.2572
40	6.6418	6.2335	5.8713	5.5482	5.2582
41	6.6450	6.2358	5.8729	5.5493	5.2590
42	6.6478	6.2377	5.8743	5.5502	5.2596
43	6.6503	6.2394	5.8755	5.5510	5.2602
44	6.6524	6.2409	5.8765	5.5517	5.2607
45	6.6543	6.2421	5.8773	5.5523	5.2611
46	6.6559	6.2432	5.8781	5.5528	5.2614
47	6.6573	6.2442	5.8787	5.5532	5.2617
48	6.6585	6.2450	5.8792	5.5536	5.2619
49	6.6596	6.2457	5.8797	5.5539	5.2621
50	6.6605	6.2463	5.8801	5.5541	5.2623

ADDITIONAL READING

Belcher MC. Lighting cost analysis: Is there a better way? Lighting Design Applic. 1989 Jun:14.

Horngren CT. Cost Accounting: A Managerial Emphasis. Englewood Cliffs, NJ: Prentice Hall; 1977.

REFERENCES

1 Helms RN, Belcher MC. Lighting for Energy Efficient Luminous Environments. Upper Saddle River, NJ: Prentice Hall; 1991.

2 Belcher MC. MCSEALS: A Monte Carlo simulation for economic analysis of lighting systems. J Illumin Engineering Soc. 1989;18(2):40.

3 Merril GS. The economics of light production with incandescent lamps. Trans ilium Eng Soc. 1937;XXXII:1007.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________
   Affiliation: ______________________________
   Address: ______________________________
   City: ____________ State: ________ Zip: ________ Country: ____________
   Telephone: ______________________________
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   E-mail: ______________________________

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: ______________________________ Date: ____________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
   [ ] Change to read as follows
   [ ] Delete and substitute as follows
   [ ] Add new text as follows
   [ ] Delete without substitution

Use underscore to show material to be added (<u>added</u>) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
   [ ] Additional pages are attached. Number of additional pages:
   [ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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